3

Biomolecules in solution

THE most common reactions happen in solution. Think, for example, when you add sugar to your coffee or how a metal rusts when it gets wet. The first example involves a dissociation reaction, whereas the second a redox reaction. This chapter first covers the properties of solutions and the quantification of the amount of solute in a solution. This chapter also covers some important types of reactions happening in solution. Another important concept covered in this chapter is the idea of electrolytes. On one hand, most of you will be surprised to know that water do not conduct the electricity. This is because pure water is a weak electrolyte. On the other hand, the importance of electrolytes is well know among the sports community. If you have ever played a sport, you have probably chugged a sport drink. These are actually electrolyte solutions. However, few know the specifics of their function. Electrolytes are actually salts that conduct electricity in water by separating into positive and negative ions.

GOALS

Chemistry and molecules

1. Identify $d-$, $l-$ α or β carbohydrates
2. Break down/assemble carbohydrates

Acids and bases

1. Name and formulate ionic compounds
2. Name and formulate acids and bases

Electrolytes and solutions

1. Classify electrolytes
2. Name and formulate oxosalts and common chemicals

Reactions and energy

1. Break down/assemble nucleic acids
2. Break down/assemble peptides

The chemistry of cancer

1. Describe the dangers of radiation

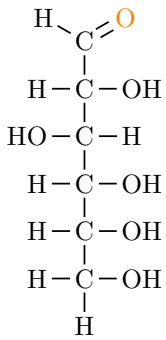
Let's calculate

1. Add and remove prefixed
2. exchange prefixes

3.1 Simple Carbohydrates

This first section introduces one of the most important biomolecules: the carbohydrates. If you think about rice, bread, and even fruits, they all contain carbohydrates as their main ingredients. This section cover the rules for their naming of single carbohydrates, called monosaccharides, and will show you how to identify these common molecules in two different forms: a line structure and a cycle, the later called Haworth structure, named after Sir Norman Haworth, a famous english biochemist. Finally carbohydrates can form dimer and polymers and these more complex molecules will also be addressed in this section.

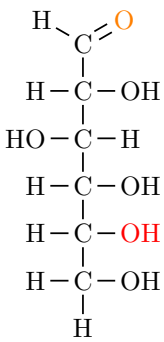
Carbs are made of C, H and O Carbohydrates are made of carbon and oxygen and hydrogen, thus their name: carbon hydrates. The simples carbohydrate is glucose with molecular formula $C_6H_{12}O_6$ and with structure



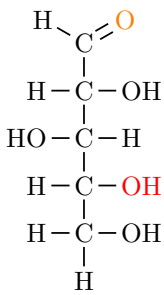
This molecule is produced in plants by means of a process called photosynthesis, in which carbon dioxide and water are added together with the help of sunlight to produce glucose and oxygen. At the same time our bodies burn glucose with oxygen while we breath.

$$6\,CO_2 + 6\,H_2O + \text{☼} \underset{\text{Respiration}}{\overset{\text{Photosynthesis}}{\rightleftharpoons}} C_6H_{12}O_6 + 6\,O_2$$

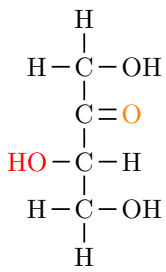
Monosaccharides: aldo or keto? The simples sugars are called monosaccharides and their name ends in *ose* like in gluc*ose*. These sugars are made of C, O and H and contain a few functional groups that you should be familiar with. They contain alcohol groups and either and aldehyde ($R-\overset{\overset{O}{||}}{C}-H$) or a ketone ($R-\overset{\overset{O}{||}}{C}-R'$) group in the carbon number one. Depending on the number of carbons, monosaccharides can be classifies in triose, tetrose, pentose, or hexose. Here a few examples:



An alcohexose



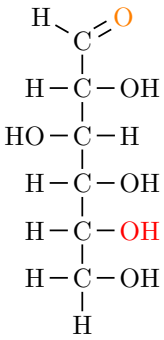
An aldopentose



A ketotetrose

Sample Problem 1

Classify the following monosaccharide as aldo or keto:

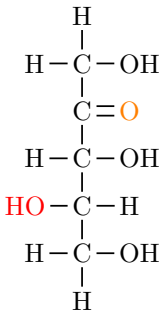


SOLUTION

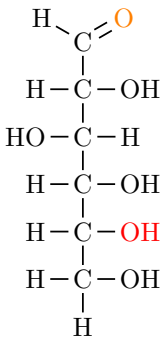
The monosaccharide has a $R-\overset{\overset{\displaystyle O}{\|}}{C}-R'$ group on carbon one and therefore it will be aldo. As it also has six carbons, it will be an aldohexose.

STUDY CHECK

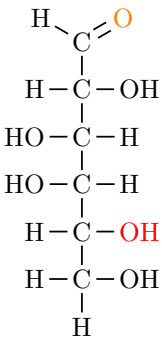
Classify the following monosaccharide as aldo or keto:



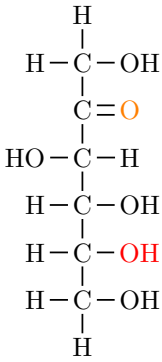
Some important monosaccharides The most important monosaccharides are glucose, fructose and galactose. These are all hexose with six carbon atoms. Glucose and galactose are aldohexose whereas fructose is a ketohexose.



Glucose



Galactose

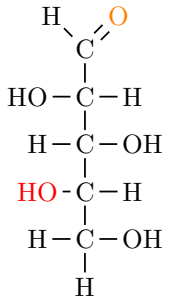


Fructose

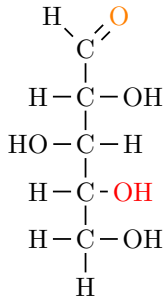
Glucose is also known as dextrose or blood sugar, is the most important simple sugar involved in the metabolism of humans. Glucose is found in corn syrup and honey as well as in fruits and vegetables in the form of more complex carbohydrates. Galactose can not be found in free form, but is found in the form of lactose, a disaccharide, in milk. Only when lactose is broken down you can find free galactose. Fructose is the sweetest of all carbs found in fruits–thus its name–and honey.

d− and l− monosaccharides Monosaccharides are classified as $d-$ and $l-$ depending on the position of the second OH group starring from the bottom of the molecule.

If that specific OH points to the right we call it $d-$ monosaccharides, whereas $l-$ have the second OH group starring from the bottom pointing left. Here some examples:



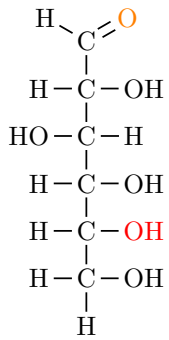
$l-$ monosaccharide



$d-$ monosaccharide

Sample Problem 2

Classify the following monosaccharide as $d-$ or $l-$:

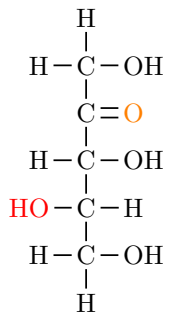


SOLUTION

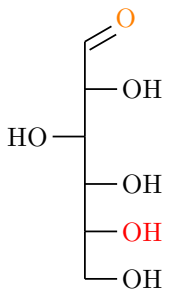
As the second OH group starting from the bottom point to the right, it will be $d-$, in particular it would be a: $d-$aldohexose.

STUDY CHECK

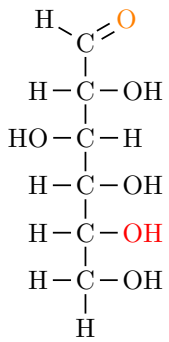
Classify the following monosaccharide as $d-$ or $l-$:



Skeletal structure of monosaccharides Often times, the structure of carbohydrates are represented in skeletal form, ignoring the C and H atoms and showing only oxygen and OH groups. Here an example:

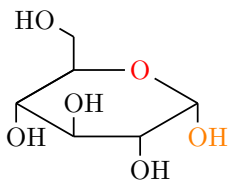


Linear Glucose (Skeletal form)

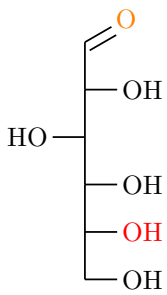


Linear Glucose (Full form)

Haworth cyclic structure of monosaccharides Normally monosaccharides exist in liquid solution. In these conditions its structure is not a linear structure such the one presented before. Instead they exist in a cyclic form called Haworth structure. The goal of this section is just to introduce you to this cyclic form. Here an example:

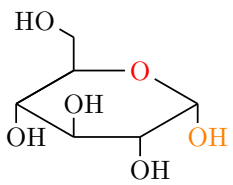


Haworth Glucose

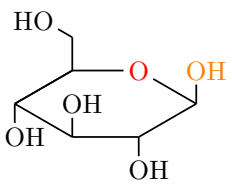


Linear Glucose (Skeletal form)

α or β form of Cyclic monosaccharides The Haworth form of monosaccharides are classified as α or β depending on the orientation of a single OH group: the OH group of the C atoms directly connected to the O atom in the cycle. If this OH groups points down we call this α and if its points up we call the structure as β . Here some examples:



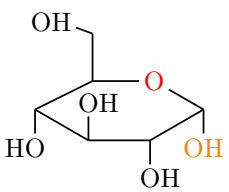
α-Glucose



β-Glucose

Sample Problem 3

Classify the following haworth monosaccharide as α or β :

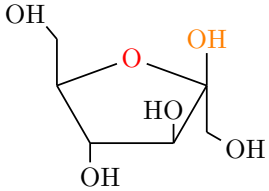


SOLUTION

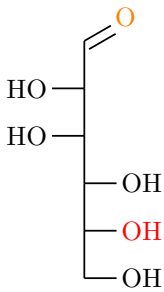
As the second OH group after the -O- in the cycle points down, it will be α.

STUDY CHECK

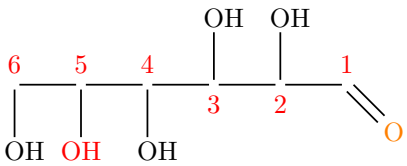
Classify the following haworth monosaccharide as α or β :



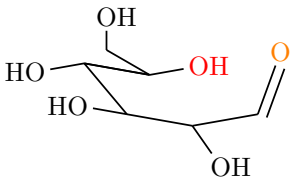
Obtaining Haworth cyclic structure of monosaccharides Linear monosaccharides cycle forming Haworth cycles. In order to obtain a cycle you just need to get the linear structure and turn in clockwise 90 degree until is horizontal, the carbonyl or keto will fall in the right part of the structure and the last C atom in the left part. After that you need to draw the structure in a boat shape, as the Oh from carbon 5 will attach carbon number one and produce a cycle.



D-Mannose

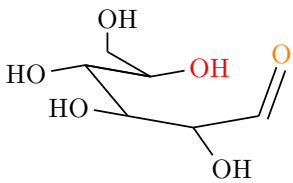


D-Mannose rotated

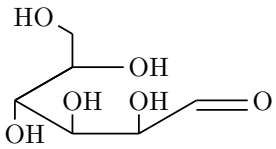


D-Mannose as a boat

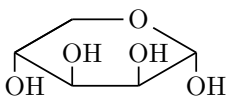
The final cycle looks like:



D-Mannose

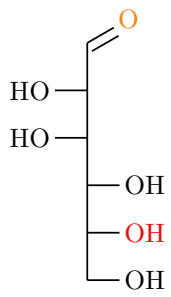


D-Mannose rotated

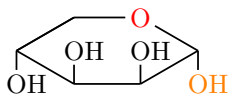


α-Mannose

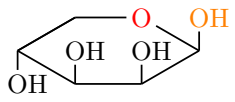
In general the cycles formed from linear monosaccharides can have five of six atoms, depending on the aldo/keto nature of the monosaccharides. Aldo monosaccharides produce six-atoms rings and keto monosaccharides produce five-atoms rings. A six-atom ring example:



D-Mannose

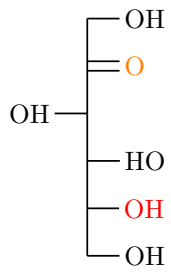


α-d-Mannose

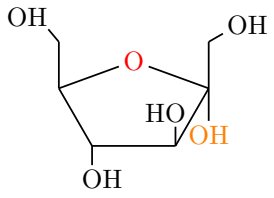


β-d-Mannose

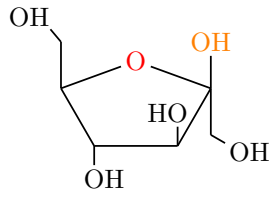
A five-atom ring example:



d-fructose



α-d-fructose

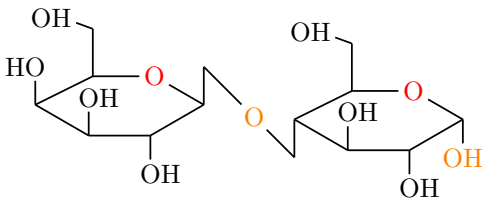


β-d-fructose

3.2 Disaccharides and polysaccharides

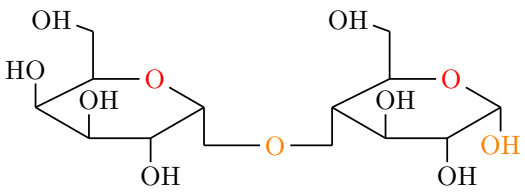
This next section covers the properties of the di and polysaccharides. These are more complex saccharides made of two of more units of sugar. At the same time, these form the most common carbs. For example, sugar cane is a disaccharides, that is, contains two monosaccharides units. Differently, the starch in potatoes is a polysaccharide, made of numerous saccharides units, and cellulose is a structural polysaccharide used by plants. All this complex saccharides have very different properties and for example humans can easily digest sugar cane and starch but not cellulose. The difference is in the way the monosaccharides are connected among themselves by means of a connection names glycosidic bond.

`Disaccharides` Disaccharides are the combination of two saccharides. An example of a disaccharides is α-lactose, found in the milk:



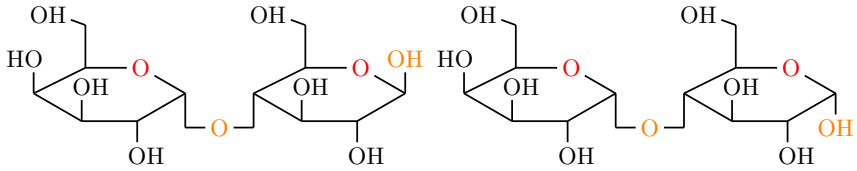
α-lactose

Another example is α-maltose, found in beer:



α-maltose

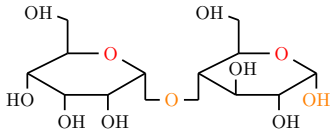
α- and β-Disaccharides Same as monosaccharaides, disaccharides can also be α- and β, depending on the position of the first OH after the -O- atom in the right cycle. For example:



β-maltose α-maltose

Sample Problem 4

Classify the following disaccharide as α or β:

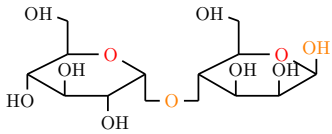


SOLUTION

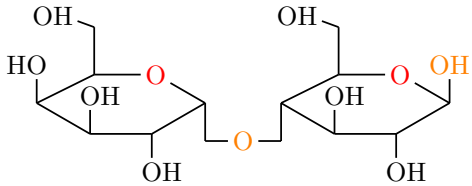
This is an α disaccharide as the first OH after the -O- atom in the right cycle is pointing down.

STUDY CHECK

Classify the following disaccharide as α or β:

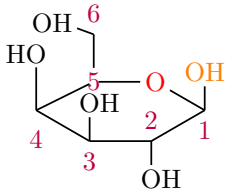


Numbering saccharides Disaccharaides are the result of the connection between two monosaccharides at a specific location of the molecule. Depending on the location of the connection and the orientation of the connection, the disaccharide would be easy or impossible to digest. Is is important to number the carbons in each monosaccharide in order to be able to identify the atoms involved in the connection. Here is an example of a monosaccharide:

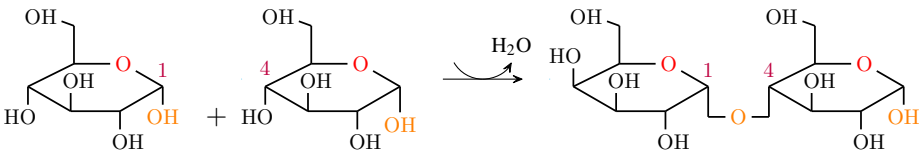


β-maltose

In order to number the chain we start by the -O- atom. This atoms would be zero. After that you continue numbering clockwise. The results is:



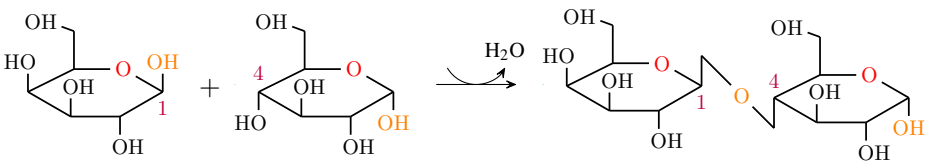
`Glycosidic bond` Monosaccharides connect to form disaccharides by means of very special bond. This bond is the result of combining two OH groups and looks like -O-. This is the glycosidic bond and the formation of this bond produced water. For example, maltose is the result of the connection of two glucose molecules by means of a 1-4-glycosidic bond.



α-glucose and α-glucose

Maltose (α-1-4-Glycosidic bond)

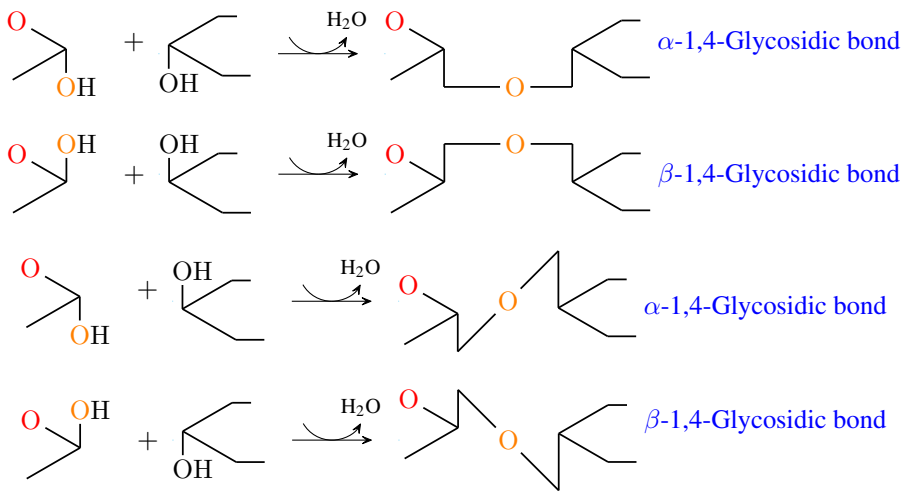
So if you combine an α-glucose molecule with another α-glucose molecule you form a 1-4-glycosidic bond. As the first glucose is α the resulting bond will also be an α bond, and the full name for the bond will be α-1-4-glycosidic bond. Now, depending on the orientation of the OH groups involved in the glycosidic bond we can have four different orientations of the bond. For example, lactose results from combining β-D-galactose and α-D-glucose. The separate monomers are:



β-galactose and α-glucose

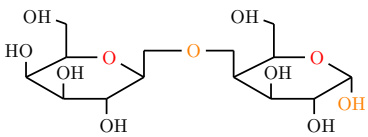
Lactose (β-1-4-Glycosidic bond)

In general the four possible glycosidic bonds are:



Sample Problem 5

Classify the following glycosidic bond:

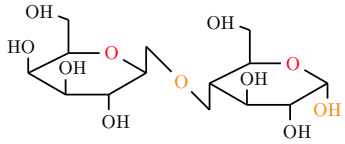


SOLUTION

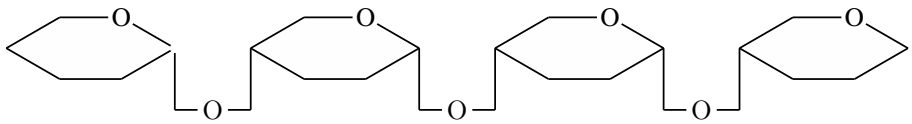
This is a β-1-4-glycosidic bond, as the first bond in the glycosidic bond points up.

STUDY CHECK

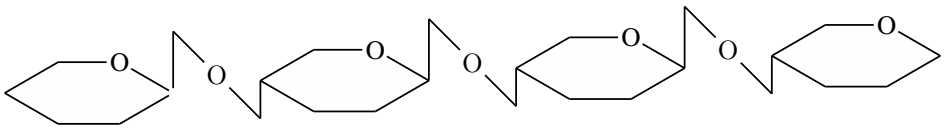
Classify the following glycosidic bond:



`Polysaccharides: starch and cellulose` In the previous section you saw the structure of maltose and lactose and these were made of monosaccharides connected by means of a -O- bond called glycosydic bond. A polysaccharide is a long polymer formed by means of numerous units of sugar connected by means of glycosidic bonds. Starch for example is a polysaccharide found in plants, rice, wheat or potatoes is made of α-glucose molecules connected by means of a α-1-4Glycosidic bond. These bonds are easy to break and humans easily digest starch. Starch hydrolyze in water and acid ultimately producing glucose.

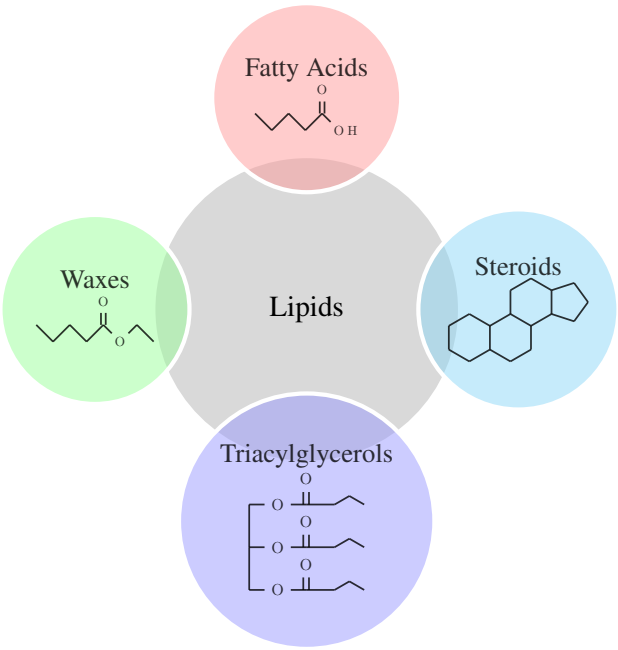


Cellulose is also a polymer of α-glucose molecules. However, this polysaccharide contains glucose molecules bonded by means of a β-1-4Glycosidic bond. These bonds are stronger than the α-1-4Glycosidic bond, and hence humans can not digest cellulose. Only animals can break this polymer with the help of a specific bacteria.

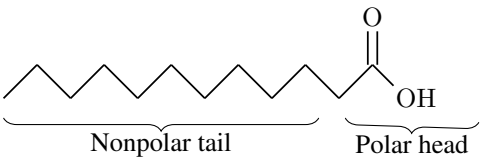


3.3 Lipids

This section covers the chemical properties and structure of lipids. These biomolecules are often referred as fats as they are insoluble in water and soluble in organic solvents. Think about oil, butter or even cholesterol. All these are different types of lipids with distinct structure. One one hand, oil and butter contain fatty acids, which are organic acids–carboxylic acids–with a carboxylic group connected to a very long zigzag hydrocarbon chain. On the other hand, cholesterol is a lipid that do not contain fatty acids. The structure of cholesterol results from an steroid nucleus made of three cyclohexane and a cyclopentane fused together. Perhaps less known fats are waxes and triacylglicerols. Both result from the reaction of fatty acids and alcohols. We will briefly cover the structure of those lipids as well. The following diagrams gives the classification for the different kinds of lipids.



Structure of fatty acids Fatty acids are molecules with two different parts: a polar head and a nonpolar tail. On one hand, a carboxylic acid group ($\overset{\overset{\displaystyle O}{||}}{C}-OH$) make these molecules acidic and polar. On the other hand, a hydrocarbon ( ) chain makes these molecules nonpolar, that means and hence insoluble in water. As the chain tends to be long these molecules over all are nonpolar that means they do not dissolve in water. These fatty acids are called saturated fatty acids, as they only contain simple C-C bonds.



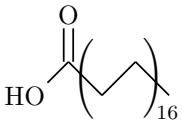
Saturated Fatty acid

Sample Problem 6

Stearic acid is a saturated fatty acid with 18 carbons found in animal fat. Write down its skeletal formula.

SOLUTION

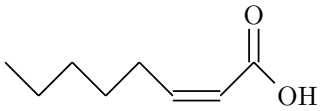
Saturated fatty acids have a polar acid head and a hydrocarbon chain with no double bonds. The polar head has one carbon and hence the tail should have 17 carbons:



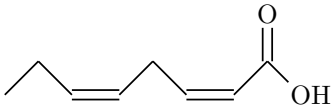
STUDY CHECK

Lignoceric acid is a saturated fatty acid with 24 carbons found in wood tar and peanut oil contain.Write down its skeletal formula.

Mono and polyunsaturated fatty acids Often times fatty acids contains double bonds called unsaturations. They can have a single double bond or more than one double bond. Monounsaturated fatty acids only contain a single unsaturation and polyunsaturated fatty acids have more than one double bond. Examples are palmitoleic acid, a monounsaturated fatty acid found in butter or linolenic acid, a polyunsaturated fatty acid found in corn oil.



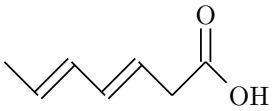
Monounsaturated fatty Acid



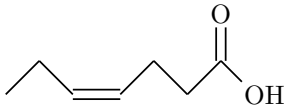
Polyunsaturated fatty Acid

Sample Problem 7

Classify the following fatty acids as saturated, monounsaturated or polyunsaturated:



A



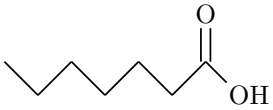
B

SOLUTION

A saturated fatty acid has no double bond on its tail, whereas a monounsaturated fatty acid has a single double bond in the tail and a polyunsaturated fatty acid has more than two double bonds. According to this, the fatty acid A is a polyunsaturated as it contains two double bonds on its tail. The fatty acid A is monounsaturated, as it only contains a single double bond.

STUDY CHECK

Classify the following fatty acid as saturated, monounsaturated or polyunsaturated:



Melting point of fats Some fats such as olive oil are liquid at room temperature. Other such as butter or lard are solid at room temperature. Liquid fats have low melting point so that they are already melted in a liquid form at room temperature. Solid fats have in general larger melting point and hence at room temperature they preserve its solid form. Two different factors affect the melting point of a fat. The first factor is the number of carbons. The larger the number of carbons the higher melting point and hence the fat will most likely be solid at room temperature. For example, lauric acic, found in coconut oil, has 12 carbon atoms and a boiling point of 44°C. Palmitic acid, an oil found in palm, on the other hand has 16 carbon atoms and its boiling point would be therefore higher (63°C).

$$CH_3-\left(CH_2\right)_{12}-COOH$$

Lauric Acid (MP=44°C)

$$CH_3-\left(CH_2\right)_{14}-COOH$$

Palmitic Acid (MP=63°C)

The second factor is the presence of double bonds–remember these are called unsaturations. Double bonds decrease the melting point. For example, stearic acid is an unsaturated fatty acid–with no double bonds–with 18 carbons and a melting point of 69°C. On the other hand, linoleic acid is a monounsaturated fatty acid with the same 18 carbons but a lower melting point of -5°C. Due to the presence of a single unsaturation the melting point is highly reduced and hence linoleic acid will be liquid at room temperature whereas stearic acid will be a solid.

$$CH_3-(CH_2)_4-CH=CH-CH_2-CH=CH-(CH_2)_7-COOH$$

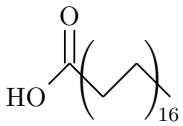
Linoleic Acid (MP=−5°C)

$$CH_3-(CH_2)_{16}-COOH$$

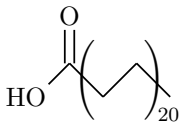
Stearic Acid (MP=69°C)

Sample Problem 8

Compare the melting point of the following fatty acids and predict whether they will likely be solid or liquid at room temperature.



A



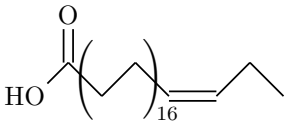
B

SOLUTION

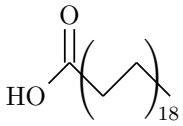
The larger the number of atoms in the fatty acid the higher its melting point and hence the higher the chances the fat is solid at RT. The fatty acid A has 33 carbon atoms whereas B has 41 so B has higher melting point than A and has more chances to be solid at RT.

STUDY CHECK

Compare the melting point of the following fatty acids and predict whether they will likely be solid or liquid at room temperature.

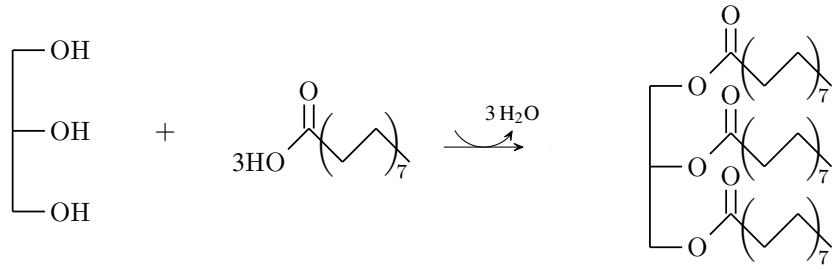


A



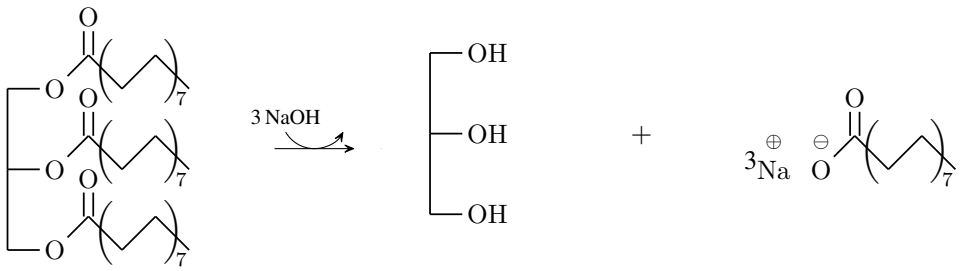
B

Waxes, animal fat and soaps Waxes and animal fat are the results of a reaction between fatty acids and alcohols. Waxes are ester of a fatty acid produced by mixing a fatty acid with long chain alcohols. Waxes are found in animals and plants. Plants secrete waxes in order to control evaporation, reduce wettability and increase hydration. The most important commercial plant wax is carnauba wax, a hard wax obtained from Brazilian palm employed for example in car and furniture polish, floss coating, and surfboard wax. Animal fat is also known as triacylglycerols as they result of the reaction between fatty acids and glycerol, an alcohol with three carbon atoms. Animals store energy in the form of fat. For example, during hibernation they survive by consuming fat.



Glycerol Fatty acids triacylglycerol

Soap results from the treatment of fat with a strong base such as NaOH or KOH. Due to this basic treatment fats break down into glycerol and a salt of the fatty acid. Soaps are indeed salts of fatty acids. Depending on the nature of the base employed in the saponification and the nature of the oil one can produce solid or more more liquid soaps.

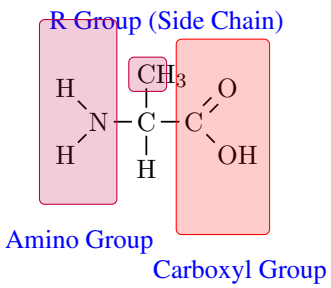


triacylglycerol glycerol Soap

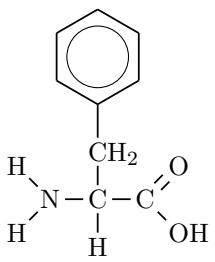
3.4 Proteins

This new section covers the chemical properties of proteins, which are polymers amino acids. Amino acids are molecules with both an amino (NH_2) and a carboxylic acid group (COOH). Amino acids have also a side chain (R)–this is a organic group different than amino or acid group. Side chains give amino acids acid-base hydrophilic-hydrophobic properties and properties. Amino acids bond together by means of a peptide amide bond producing small peptides or large proteins.

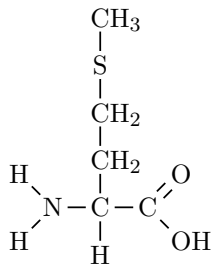
`Amino acids` Proteins are made of amino acids. Amino acids have the following parts: (1) a central carbon called α carbon; (2) an amino (NH_2) group connected to the α carbon; (3) a carboxylic acid (COOH) group connected to the α carbon; (4) a hydrogen atom (H) connected to the α carbon; (5) a side chain (R). All amino acids have the same α carbon, amino group and acid group as well as hydrogen connected to the α carbon. They differ in the side chain. Here is an example of an amino acid called Alanine, indicating the amino, the acid and the side chain.



Two more amino acids examples:



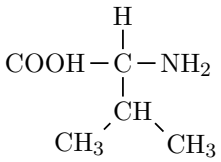
Phenylalanine



Methionine

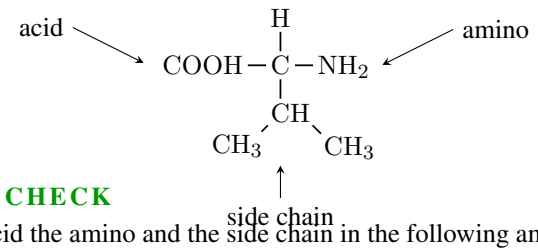
Sample Problem 9

Identify the acid the amino and the side chain in the following amino acid:



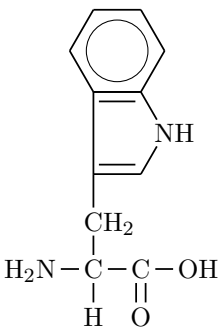
SOLUTION

The acid group is COOH and the amino is NH_2. The side chain is the other group connected to the central C different than hydrogen:

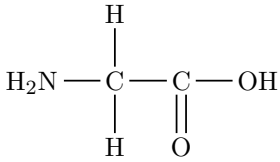


STUDY CHECK

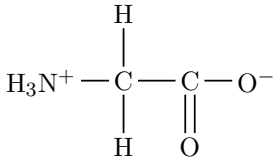
Identify the acid the amino and the side chain in the following amino acid:



`Amino acids as zwitterions` Amino acids are acid and basic, that means they have two different PK_a. For example the two different PK_a values for Glycine is 2.34 and 9.60. This means that in solution amino acids will exist in a protonated form in which the acid group looses its proton H^+ becoming negatively charged and the amino group has an extra proton and becomes positive. This positive and negative state of an amino acid is called zwitterion.



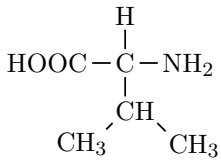
Neutral Glycine (Gly)



Glycine as zwitterion

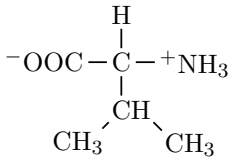
Sample Problem 10

Write down the zwitterion state of the following AA



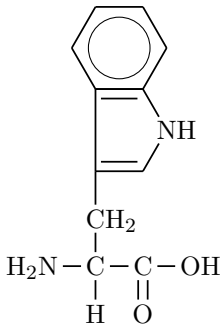
SOLUTION

In its zwitterion state an AA has a netaive acidic group and a positive amino group. The acid group has one less H and the amine gains that proton:

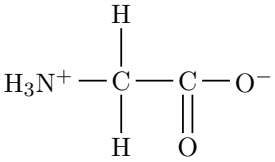


STUDY CHECK

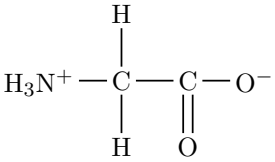
Write down the zwitterion state of the following AA



Hydrophilic and hydrophobic Amino acids There 20 common amino acids. They can be classified according to they hydrophilic and hydrophobic. Hydrophilic means that they solve in water whereas hydrophobic means that they do not solve in water. The classification is result of their side chain, as some of their side chains are hydrophilic whereas others are hydrophobic. For example, Glycine is an hydrophobic amino acid as its side chain is mainly made of carbon and hydrogen and is hence nonpolar. Differently, Serine is an hydrophilic amino acid as its side chain contains oxygen and is hence polar.

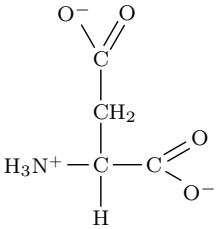


Glycine (Gly)
Hydrophobic AA

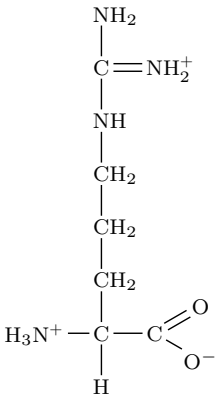


Histidine (His)
Hydrofilic AA

Acid and basic Amino acids Amino acids can also be classified according to their acid-base character. This property depends on the acid-base character of the side group. If this group is acidic then the amino acid will be acidic and basic side chains correspond to basic amino acids. For example Aspartic acid is an acidic amino acid as its side chain is acidic and arginine is a basic amino acid as its side chain is basic.



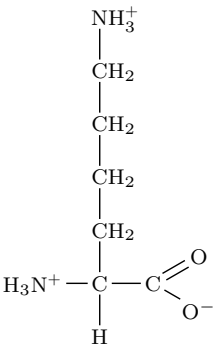
Aspartic Acid (Asp)
Acidic AA

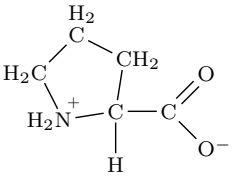


Arginine (Arg)
Basic AA

Sample Problem 11

Classify the following AA as acidic, basic, hydrophilic or hydrophobic:



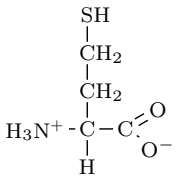


SOLUTION

They key to classify the different AA is on its side chain. The AA on the left has a nonpolar side chain made of C and H and hence it will hydrophobic. The AA on the right has a polar side chain with N and at the same time a basic side chain. This AA will be hydrophilic and at the same time basic.

STUDY CHECK

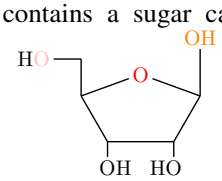
Classify the following AA as acidic, basic, hydrophilic or hydrophobic:



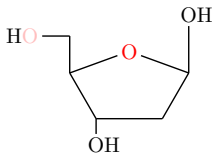
3.5 Nucleic acids

Nucleic acids are long polymers of a unit called nucleotide connected by means of a bond called phosphodiester. There are tow different types of nucleic acids: DNA and RNA. The term DNA stands for deoxyribonucleic acid, whereas the term RNA stands for ribonucleic acid. A nucleotide ies made of a phosphate acid and a nucleoside, and at the same time a nucleoside is made of a sugar and a base. This section cover the properties of nucleic acids.

Sugars in nucleic acids The structure of nucleic acids is indeed sugar. Two different sugars can be found in nucleic acids. Deoxyribonucleic acid (DNA)contains deoxyribose, hence its name, whereas ribonucleic acid (RNA) contains a sugar called ribose. The structures of ribose and deoxyribose are:



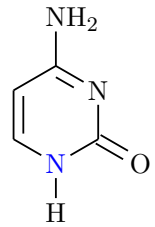
Ribose (R), RNA



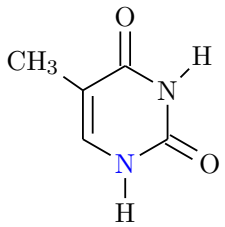
Deoxyribose (DR), DNA

Bases in DNA and RNA Nucleic acids contain a sugar, and also contain a base. Remember that bases are chemicals that produce hydroxyls in water. There are tow different types of nucleic bases: purine bases and pyrimidine bases. At the same time, there are three pyrimidine bases: Cytosine, Thymine, and Uracil. There are two purine bases: Adenine and Guanine. Some of these bases can only be found as part of DNA such as thymine, whereas others can be found only in RNA such as Uracil. Other nucleic bases can be found either on DNA or on RNA. An example of the later is Cytosine. Because of the complex name of these bases often times we refer to them by the first letter. For example, Uracil is also referred as U.

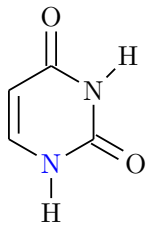
PYRIMIDINE BASES



Cytosine (C)
DNA & RNA

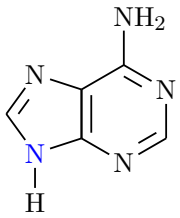


Thymine (T)
DNA

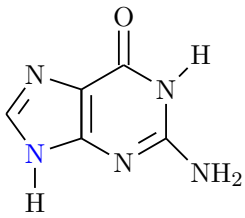


Uracil (U)
RNA

PURINE BASES

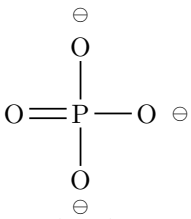


Adenine (A)
DNA & RNA

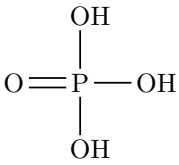


Guanine (G)
DNA

DNA and RNA contain phosphate At this point we saw tow different sugars part of RNA and DNA. We also saw some bases that also are part of the two different nucleic acids. Still, DNA and RNA are complex molecules that also contain a third element: a phosphate unit. Phosphate results from phosphoric acid: H_2PO_4 after removing two hydrogens:



Phosphate

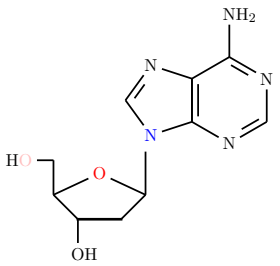


Phosphoric Acid

Nucleoside: a base connected to sugar Nucleic acids are polymers. This means the are the result of the connection between smaller units that over all make the nucleic acids. These units are called nucleosides. A nucleoside is the result of connecting a sugar (either R or DR) with a base (C, T, U, A or G). These tow units are connected by means of a bond called N-glycosidic. This is a similar bond that connect sugars in disaccharides, however involved a nitrogen instead of an oxygen:

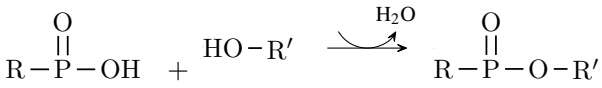
$$R-OH \quad + \; HN-R' \xrightarrow{H_2O} R-N-R' \qquad \text{N-Glycosidic bond}$$

Let us create a nucleoside. In order to do this, we need to select a sugar (for example DR) and a nucleic base (for example adenine). Both will combine be means of carbon number 1 of the sugar and eliminate a water molecule:



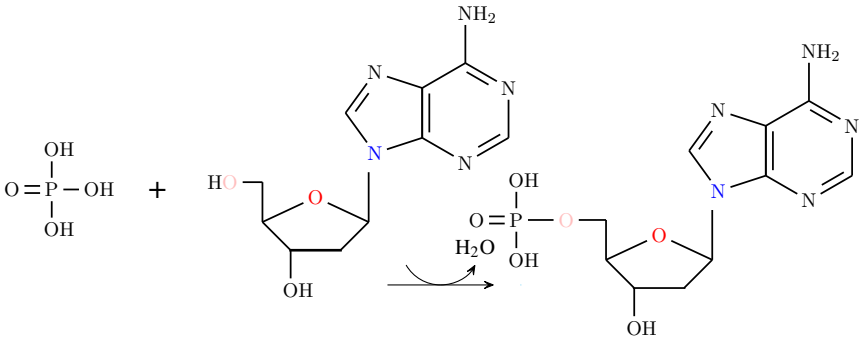
Nucleoside (Sugar+Base)

`Nucleotide: a nucleoside connected to phosphate` DNA and RNA are composed of numerous units called nucleotides. Nucleotides are the result of a nucleoside reacting with phosphoric acid by means of carbon number 5'. The bond in a nucleotide is called phosphoester bond:



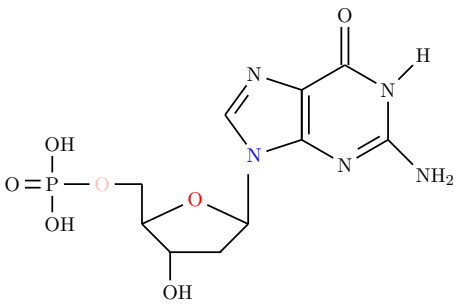
Phosphoester bond

For example, let us consider the nucleoside formed by the reaction of the Deoxyribose sugar and adenide. This nucleoside is called adenosine. After the reaction with phosphoric acid nucleotide is formed called adenosine-5'-monophosphate.

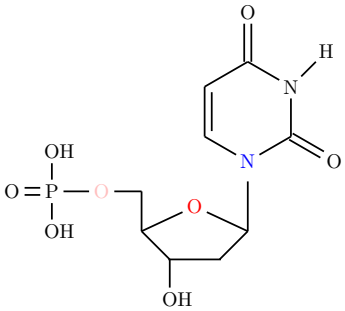


Phosphoric acid Nucleoside Nucleotide (Phosphate+Nucleosite)

Here some other examples of DNA nucleotides. Mind they are all based in Deoxyribose, the sugar that forms only DNA:

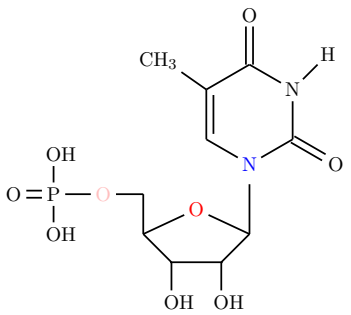


Deoxyguosine-5'-monophosphate

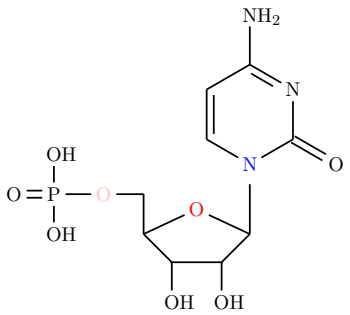


Deoxyuridine-5'-monophosphate

Here some other examples of RNA nucleotides. Mind they are all based in ribose, the sugar that forms only RNA:



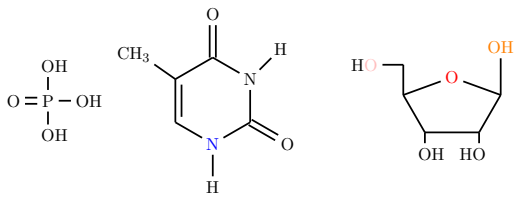
thymidine-5'-monophosphate



cytidine-5'-monophosphate

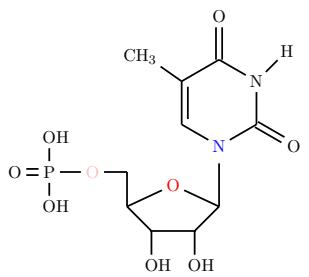
Sample Problem 12

Combine the following molecules to produce a nucleoside:



SOLUTION

A nucleoside is the result of combining a sugar–either ribose or deoxyribose–and a base. The sugar given is ribose and the base is thymine, so the nucleoside will be part of RNA:



Thymidine-5'-monophosphate

3.6 Ions & ionic charges

Atoms gain and loose electrons to produce ions. An ion is just an atom with a positive or negative charge. Ions result from an electron transfer. Positive ions have lost negatively charged electrons, whereas negative ions have gained electrons. The reason for this electron transfer is that atoms try to achieve a very stable electronic configuration with eight electrons in the valence, and this is called the octet electron configuration. Examples of ions are: H^+, Ca^{2+} or O^{2-}. This section covers the properties of ions and the ionic charges.

`Cations and anions` Atoms that loose electrons become positively charged. These ions are called cations. Example of cations are Li^+ or Mg^{2+} called lithium cation and

magnesium cation, respectively. Atoms that gain electrons become negatively charged, as electrons have negative charge. These ions are called anions. Example of anions are F^- called fluoride or N^{3-} called nitride. The way to name anions is by using the name of the element and the suffix -ide.

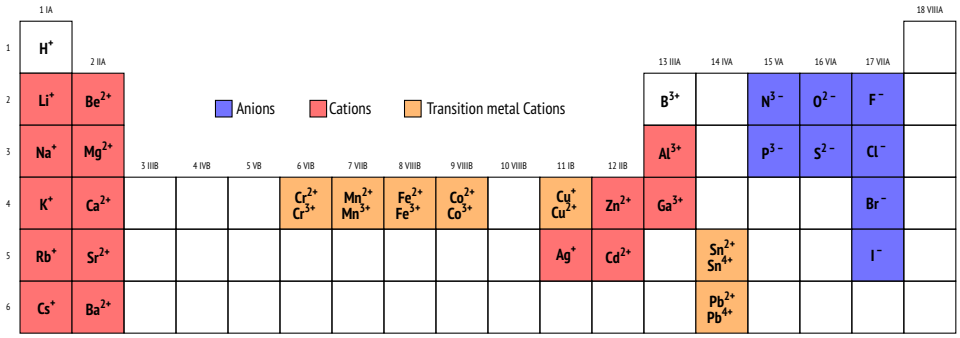


Figure 3.1 Ionic charges (valences) for different elements

Ionic charges: the valences How do we know that hydrogen produces a H^+ ion and nitrogen a N^{3-} anion. The charge of an ion is called ionic charges, and the numbers are coming from the periodic table. H, Na or K are in the group IA (left of the table) and hence the ionic charge will be $1+$. Similarly, Mg or Ca are in the group IIA (left of the table) and hence the ionic charge will be $2+$. Differently, F, Cl or Br are in the group 7A (right of the table) and its charge will be $1-$. Oxygen is in group 6A (right of the table) and the ionic charge will be $2-$. Figure 3.1 contains all ionic charges. What if the element is not in this list such as the case of Iron (Fe)? In that case, very probably it will have several ionic charges and this charge has to be indicated in the chemical name. An example would be Fe, which ionic charge is not in Figure **??** as iron can have several ionic charges.

Sample Problem 13

Identify the correct ionic state of: Cl, K, O and C.

SOLUTION

Cl is on the 7A group and hence its charge is $1-$, whereas potassium belongs to 1A and its charge will be $1+$. Oxygen and carbon will have $2-$ and $4-$ charges. The final ionic states are: Cl^-, K^+, O^{2-} and C^{4-}.

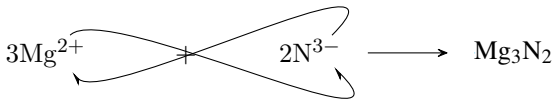
STUDY CHECK

Identify the correct ionic state of: N and Br.

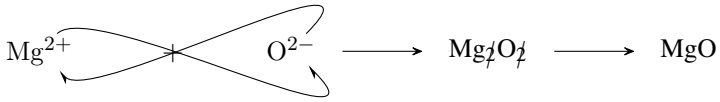
3.7 Ionic compounds

Ionic compounds are chemicals resulting from the combination of a nonmetallic element with a metallic element. And example is NaCl, which results of combining sodium (a metal) with chloride (a non metal).

Combining ions Ionic compounds are the result of combining two ions: a positive (cation) and a negative (anion) ions. Each ion has a charge, depending on its location on the table. When combining two atoms you first need to arrange the ions starting from positive and followed by negative. The charges of an ion would become the coefficient of the other ion. For example Mg^{2+} and N^{3-} are combined as Mg_3N_2:



Another example would be the combination of Na^+ and O^{2-} that would be Na_2O. You need to simplify the indexes of the formula by diving by the smallest one, always using integer values. For example, Mg^{2+} and O^{2-} give Mg_2O_2 that should be written as MgO



Another example that involves simplifying the formula is the chemical resulting of combining Ca^{2+} and C^{4-}. After combining the charges we obtain Ca_4C_2 that needs to be simplified dividing by the smallest number leading to Ca_2C.

Sample Problem 14

Combine the following ions or give the ions given the final compound:

Ions	Combination
Li^+ and O^{2-}	
Ca^{2+} and O^{2-}	
	Li_3N
	Mg_2C

SOLUTION

The result of combining Li^+ and O^{2-} is Li_2O. For Ca^{2+} and O^{2-}, the resulting chemical is CaO. Li_3N results from the combination of Li^+ and N^{3-}, and Mg_2C results from Mg^{2+} and C^{4-}.

STUDY CHECK

Combine the following ions or give the ions given the final compound: Na^+ and F^- and Na_3N.

`Simple ionic naming (type I ionic)` Type I ionic compounds result from the combination of a metal with given valence (Li, Ca, Mg, etc.) and a non metal. In order to name an ionic compound (type I ionic) you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix *-ide*. As an example, the formula NaCl is named as sodium chloride and $MgCl_2$ is named magnesium chloride. Another example would be:

$CaCl_2$	calcium chloride	LiO_2	lithium oxide

In order to formulate an ionic compound based on a name, we need to combine both ions by exchanging the valences (the ionic charges). For example,$MgCl_2$ results of the combination of Mg^{2+} and Cl^- so that the number 2 in $MgCl_2$ near the Cl atom is coming from the Mg^{2+}. In other words:

$$Mg^{2+} \quad 2Cl^{-} \longrightarrow MgCl_2$$

The sign of the charges only indicate which element goes first in the formula: the positive element (cation) first following by the negative element (anion). For example the result of combining Na^+ and Cl^- is NaCl and not ClNa as Na has positive ionic charge and has to appear first in the formula.

Sample Problem 15

Name or give the formula for the following ionic compounds:

Formula	Name
MgO	
Mg_3N_2	
	Lithium nitride
	Magnesium carbide

SOLUTION

The name for MgO is magnesium oxide. Mg_3N_2 is called magnesium nitride. The formula for Lithium nitride is Li_3N and the formula for Magnesium carbide is Mg_2C, result of simplifying Mg_4C_2 dividing by two, the smallest number.

STUDY CHECK

Name or give the formula for the following ionic compounds: Sodium fluoride and Na_3N.

3.8 Naming acids & bases

In this section we will learn how to name acids and bases. Acids normally have common names (e.g. sulfuric acid) and its naming does not follow modern rules. Names and formulas of acids are listed in tables. Differently, bases (e.g. sodium hydroxide) are named in a standard way.

`Bases or hydroxides` Bases (hydroxides) result from the combination of a metal and the hydroxide anion (OH^-). Examples are NaOH or $Ca(OH)_2$. The name of a base starts by the name of the cation finishing by the word *hydroxide*. An example is NaOH named as *sodium hydroxide*, or $Ca(OH)_2$, named as *calcium hydroxide*. The word *hydroxide* refers to the OH^- ion, and hence Sodium hydroxide results from combining Na^+ and OH^-, and Calcium hydroxide from combining Ca^{2+} and OH^-. More examples of hydroxides:

$Mg(OH)_2$	Magnesium hydroxide

Table 3.1 Names of oxoacids and oxosalts (top table) and hyracids (bottom table). Yellow rows need to be remembered.

Element	Oxoacid	Oxoacid Name	Oxoasalt	Oxoasalt Name
Manganese	$HMnO_4$	Permanganic Acid	MnO_4^{-}	Permanganate
	H_2MnO_4	Manganic acid	MnO_4^{-2}	Manganate
Carbon	H_2CO_3	Carbonic Acid	CO_3^{-2}	Carbonate
Nitrogen	HNO_3	Nitric Acid	NO_3^{-}	Nitrate
	HNO_2	Nitrous Acid	NO_2^{-}	Nitrite
Phosphorus	H_3PO_4	Phosphoric Acid	PO_4^{-3}	Phosphate
Sulfur	H_2SO_4	Sulfuric Acid	SO_4^{-2}	Sulfate
	H_2SO_3	Sulfurous Acid	SO_3^{-2}	Sulfite
	$H_2S_2O_2$	Thiosulfurous Acid	$S_2O_2^{-2}$	Thiosulfite
	$H_2S_2O_3$	Thiosulfuric Acid	$S_2O_3^{-2}$	Thiosulfate
	$H_2S_2O_7$	Disulfuric acid	$S_2O_7^{-2}$	Disulfate
	$H_2S_2O_8$	Peroxodisulfuric acid	$S_2O_8^{-2}$	Peroxodisulfate
Chlorine	$HClO_4$	Perchloric Acid	ClO_4^{-}	Perchlorate
	$HClO_3$	Chloric acid	ClO_3^{-}	Chlorate
	$HClO_2$	Chlorous acid	ClO_2^{-}	Chlorite
	$HClO$	Hypochlorous acid	ClO^{-}	Hypochlorite
Iodine	HIO_4	Periodic Acid	IO_4^{-}	Periodate
Chromium	H_2CrO_4	Chromic acid	CrO_4^{2-}	Chromate
	$H_2Cr_2O_7$	Dichromic acid	$Cr_2O_7^{2-}$	Dichromate
Boron	H_3BO_3	Boric acid	BO_3^{3-}	Borate

Hydracid	Hydracid Name	Hydracid	Hydracid Name
HCl	Hydrochloric acid	HBr	Hydrobromic acid
HI	Hydroiodic acid	HF	Hydrofluoric acid
HCN	Hydrocyanic acid	H_2S	Hydrosulfuric acid

`Acids` Acids–in particular inorganic acids–are chemicals that normally contain hydrogen at the beginning of its formula. For example, HCl or H_2SO_4. HCl is an hydracid and is named as *hydrochloric acid*, whereas H_2SO_4 is an oxoacid that contains oxygen named as *sulfuric acid*. The names of acids are not standard and they come from common names employed in the field for many years. Table 3.1 contains a list of the most important oxoacids and hydracids. More examples of acids:

HNO_3	Nitric acid	HF	Hydrofluoric acid

Sample Problem 16

Name or give the formula for the following acids and bases. Indicate whether the compound is an acid or a base.

Formula	Acid/Base	Name
HCN		
KOH		
		Carbonic acid
		Lithium hydroxide

SOLUTION

HCN is an acid named hydrocyanic acid. KOH is a base called potassium hydroxide. The formula for Carbonic acid is H_2CO_3, and Lithium hydroxide is a base with formula LiOH.

STUDY CHECK

Name or give the formula for the following ionic compounds: phosphoric acid and $Mg(OH)_2$.

3.9 Naming complex salts

At this point we saw the naming and formulation of ionic (e.g. NaCl) and covalent compounds (e.g. CO_2). This section covers the naming of complex salts: oxosalts and hydrosalts. In general, salts (oxosalts or hydrosalts) are the result of mixing an oxoacid and a base. They tend to look more complex than simple ionic or covalent compounds as they have at least three different elements. An example of oxosalt would be $CaSO_4$ called calcium carbonate. An example of hydrosalt would be $NaHSO_4$ which is called sodium monohydrosulfate. This section will also cover the naming of hydrates (e.g. $CaSO_4 \cdot H_2O$), that are compounds containing water molecules inside its structure. Before being able to name these complex chemicals it is convenient to practice combining ions, without paying attention to the naming.

Combining ions In order to combine two ions, you first arrange the positive ion in the left followed by the negative ion in the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. The positive and negative charges are not carried. If the ions have more than one element we have to use parenthesis. An example would be combining Ca^{2+} and $PO_4{}^{3-}$ leading to $Ca_3(PO_4)_2$:

$$3Ca^{2+} \quad 2PO_4^{3-} \longrightarrow Ca_3(PO_4)_2$$

We would simplify in case the charges compensate with each other.

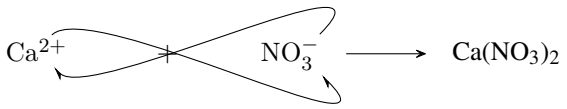
An example would be combining Mg^{2+} and $SO_4{}^{2-}$ leading to $MgSO_4$

$$Mg^{2+} \quad SO_4^{2-} \longrightarrow Mg_{\cancel{2}}(SO_4)_{\cancel{2}} \longrightarrow MgSO_4$$

Naming Oxosalts The names of the oxosalts are constructed by combining the name of the first element–you need to specify its charge in the case of a transition metal element with different possible charges–followed by the name of the oxosalt from Table 3.1. For example, the name of $MgSO_4$ is magnesium sulfate, as Mg^{2+} is magnesium and $SO_4{}^{2-}$ is sulfate. Another example is$Fe_2(CO_3)_3$ called Iron(III) carbonate. A final example:

$NO_3{}^-$	Nitrate	$LiNO_3$	Litium nitrate (oxosalt)

Formulating Oxosalts In the case that you know the name of an oxosalt and you want to obtain its formula, you first need to arrange the positive ion in the left followed by the negative ion in the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. For example, calcium nitrate results from the combination of Ca^{2+} calcium and $NO_3{}^-$, nitrate. By combining the two ions we obtain the final formula as $Ca(NO_3)_2$:



Sample Problem 17

Name of give the name of the following oxosalts:

Formula	Name
K_2SO_4	
Na_2CO_3	
	Magnesium carbonate
	Sodium phosphate

SOLUTION

K_2SO_4 is named potassium sulfate, as K^+ is potassium and $SO_4{}^{2-}$ stands for sulfate. Na_2CO_3 is sodium carbonate. Magnesium carbonate is $MgCO_3$ and sodium phosphate is Na_3PO_4.

STUDY CHECK

Name of give the name of the following oxosalts: $CaSO_4$ and Aluminum sulfate.

3.10 Electrolytes

Strong electrolytes Strong electrolytes completely dissociate in water. Hence, in a solution of a strong electrolyte you will only have ions and never molecules. Strong electrolytes are typically ionic compounds such as $MgCl_2$ or NaCl (table salt). We represent the dissociation of a strong electrolyte with a single arrow, meaning that the reaction proceeds to completion and for the example below, in the solution we will only have ions ($Mg^{2+}_{(aq)} + 2\,Cl^-_{(aq)}$) and not molecules ($MgCl_{2(s)}$):

$$MgCl_{2(s)} \xrightarrow{H_2O} Mg^{2+}_{(aq)} + 2\,Cl^-_{(aq)}.$$

Weak electrolytes Weak electrolytes partially dissociate in water, and this is indicated by means of a chemical reaction with a double arrow. Hence in a solution of a weak electrolyte you will have ions as well as molecules at the same time. Examples of weak electrolytes are hydrofluoric acid, water, ammonia or acetic acid. The dissociation of hydrochloric acid (HF) proceeds as:

$$HF_{(l)} \overset{H_2O}{\rightleftharpoons} H^+_{(aq)} + F^-_{(aq)}$$

Acetic acid (CH_3COOH) is an important weak electrolyte and its dissociation proceeds somehow in a peculiar way:

$$CH_3COOH_{(l)} \overset{H_2O}{\rightleftharpoons} CH_3COO^-_{(aq)} + H^+_{(aq)}$$

Nonelectrolytes Nonelectrolytes do not dissociate in water. Hence a solution of a nonelectrolyte will only contains molecules and not ions. Examples of nonelectrolytes are carbon-based chemicals such as methanol, ethanol, urea or sucrose. The dissociation of urea for example CH_4N_2O proceeds as:

$$CH_4N_2O_{(s)} \xrightarrow{H_2O} CH_4N_2O_{(aq)}$$

Identify the electrolyte character of a chemical You can use the Table below to identify the electrolyte character of a chemical. Ionic compounds are in general strong electrolytes, and most acids as well. There is four important weak electrolytes: water, acetic acid, ammonia and hydrofluoric acid. Covalent compounds are in general nonelectrolytes. Organic compounds, compounds based on carbon atoms (e.g. $C_{12}H_{22}O_{11}$) are in general nonelectrolytes.

Table 3.2 Different types of electrolytess

Electrolyte Type	Dissociation	Particles in solution	Examples
Strong	Fully	Mostly ions	Ionic Compounds and most acids and bases (hydroxides): NaCl, NaOH, HCl, $MgCl_2$, H_2SO_4, etc
Weak	Partially	Ions & molecules	NH_3, CH_3COOH (acetic acid), HF, H_2O
Nonelectrolytes	No	molecules	Most covalent compounds: CH_3OH (methanol), CH_3CH_2OH (ethanol), $C_{12}H_{22}O_{11}$ (sucrose), CH_4NO_2 (urea)

Sample Problem 18

For the following chemicals indicate whether you will have in the solution (a) only ions, (b) ions and some molecules, or (c) molecules: NH_3, KOH, and $C_{12}H_{22}O_{11}$.

SOLUTION

Ammonia (NH_3) is a weak electrolyte and a solution of ammonia will contain ions and well as ammonia molecules. Differently KOH is a strong electrolyte and in solution you would find only ions (K^+ and OH^-). Sucrose ($C_{12}H_{22}O_{11}$) is a nonelectrolyte and in solution you will find molecules.

STUDY CHECK

For the following chemicals indicate whether you will have in the solution only ions, ions and some molecules, or molecules: (a) H_2SO_4, HNO_3 and CH_3OH.

3.11 Radiation protection

Radioactivity results from the emission of very energetic and small particles. It can be extremely harmful when no proper protection is used. Therefore, all hospital personnel working with radioactive isotopes–radiologist, doctor, and nurse–need to be protected against radiation.

alpha particles Alpha radiation is made of very heavy particles (He nuclei) that can only travel between 2-4cm in the air before disappearing Inside your body they can penetrate only 0.05mm. A simple piece of thin clothing, a lab coat, gloves or even our skin can protect us against alpha particles.

`beta particles` Beta radiation is made of lighter particles (electrons) that move much faster than alpha particles. Beta particles travel between 200-300cm in the air and between 4-5mm in body tissue. Heavy clothing such as lab coats or gloves is needed to protect you against this radiation.

`gamma particles` Gamma radiation can pass through many materials such including body tissues. Gamma rays travel around 500 m in the air, and more than 50cm in tissue. Only very dense shielding, such as lead or concrete, will protect you from this radiation.

Table 3.3 Radiation protection

α **particle**	Travels 2-4cm in air
	Travels 0.05mm in tissue
	protected with thin clothing
β **particle**	Travels 200-300cm in air
	Travels 4-5mm in tissue
	protected with heavy clothing
γ **particle**	Travels 500m in air
	Travels 50cm in tissue
	protected with lead or concrete

3.12 Using conversion factors to switch prefixes

In order to switch a prefix into another prefix, such as transforming 30 millimeters (30 mm) into centimeters (cm), you will need two different conversion factors: the first conversion factor will remove the original unit (mm) introducing an intermediate unit, meters (m), whereas the second conversion factor will remove the intermediate meter and introduce the final unit (cm). You will get the conversion factors from Table **??**. You will arrange the first conversion factor so that the original unit cancels out with the bottom of the first conversion factor, giving you an intermediate unit. You will arrange the second conversion factor so that the intermediate unit cancels out with the bottom of the second conversion factor giving the final unit. For this example:

$$30\cancel{mm} \times \frac{1 \times 10^{-3}\cancel{m}}{1\cancel{mm}} \times \frac{1cm}{1 \times 10^{-2}\cancel{m}} = 3cm$$

Sample Problem 19

The length of a textbook page is 20cm. How many mm correspond this length.

SOLUTION

We want to convert 20 cm into mm, that is, we are switching prefixed. In order to do this, you need two conversion factors: $\frac{1 \times 10^{-2}m}{1cm}$ and $\frac{1 \times 10^{-3}m}{1mm}$. You will have to arrange the number (20cm) and the two conversion factors in the following form:

$$20\cancel{cm} \times \frac{1 \times 10^{-2}\cancel{m}}{1\cancel{cm}} \times \frac{1mm}{1 \times 10^{-3}\cancel{m}} = 200mm$$

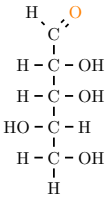
STUDY CHECK

Convert 100mm to km.

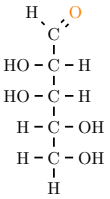
CHAPTER 3

SIMPLE CARBOHYDRATES

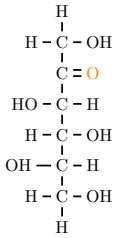
3.1 Classify the following carbohydrate as aldo or keto



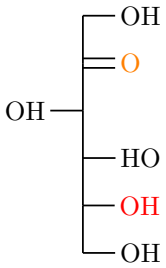
3.2 Classify the following carbohydrate as aldo or keto



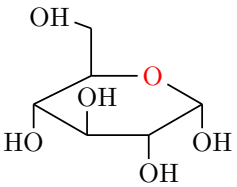
3.3 Classify the following carbohydrate as aldo or keto



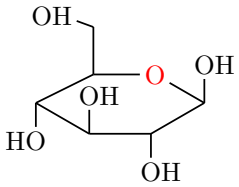
3.4 Find the molecular formula of the following monosaccharide:



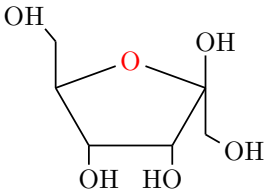
3.5 Classify the following Haworth monosaccharide as α or β.



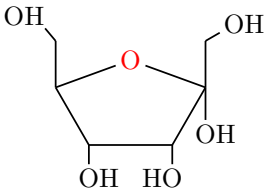
3.6 Classify the following Haworth monosaccharide as α or β.



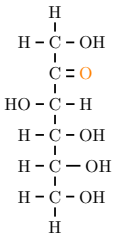
3.7 Classify the following Haworth monosaccharide as α or β.



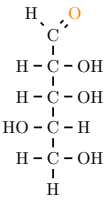
3.8 Classify the following Haworth monosaccharide as α or β.



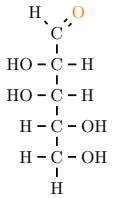
3.9 Classify the following monosaccharide as d or l.



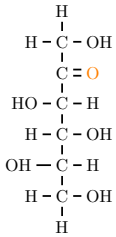
3.10 Classify the following monosaccharide as d or l.



3.11 Classify the following monosaccharide as d or l.

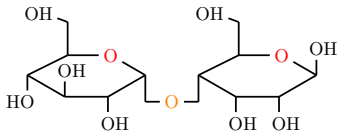


3.12 Classify the following monosaccharide as d or l.

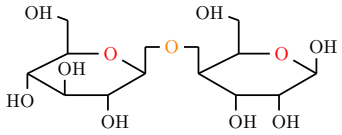


DISACCHARIDES AND POLYSACCHARIDES

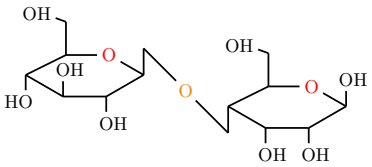
3.13 Classify the following glycosidic bond.



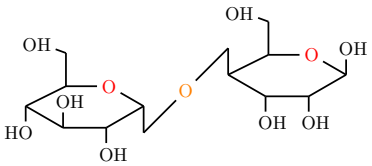
3.14 Classify the following glycosidic bond.



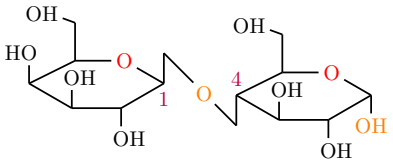
3.15 Classify the following glycosidic bond.



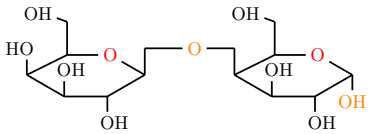
3.16 Classify the following glycosidic bond.

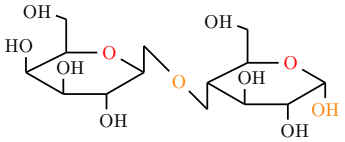


3.17 Enzymes that split disaccharides are only found in the upper reaches of the small intestine (called the duodenum). Find the monosaccharides resulting from the digestion of the lactose:

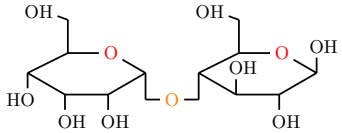


3.18 Which of the following disaccharides is easier to digest:

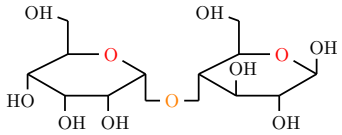




3.19 Classify the following disaccharide as α or β.

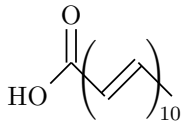


3.20 Classify the following disaccharide as α or β.

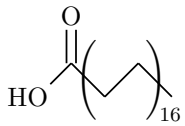


LIPIDS

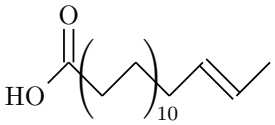
3.21 Classify the following fatty acid as saturated, monounsaturated and polyunsaturated:



3.22 Classify the following fatty acid as saturated, monounsaturated and polyunsaturated:

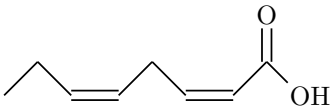


3.23 Classify the following fatty acid as saturated, monounsaturated and polyunsaturated:

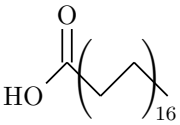


3.24 Give the molecular formula of the following fatty acid:

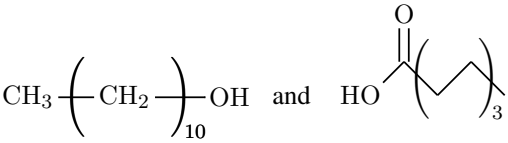
3.25 Give the molecular formula of the following fatty acid:



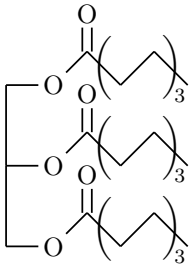
3.26 Give the molecular formula of the following fatty acid:



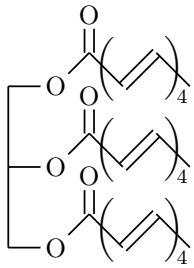
3.27 Give the structure of the wax resulting of mixing the following fatty acid and alcohol:



3.28 Give the structure of the soap resulting of saponification of the following fat:

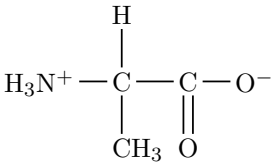


3.29 Give the structure of the fatty acid used to form the following fat:

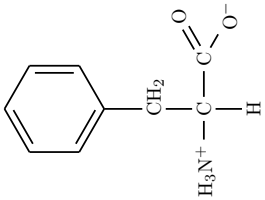


Proteins

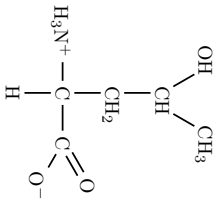
3.30 Identify the side chain of the following amino acid:



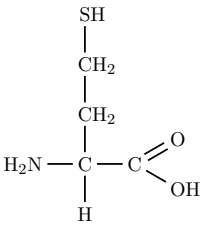
3.31 Identify the side chain of the following amino acid:



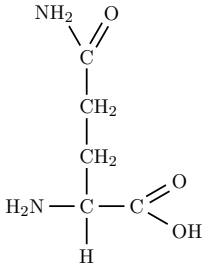
3.32 Identify the side chain of the following amino acid:



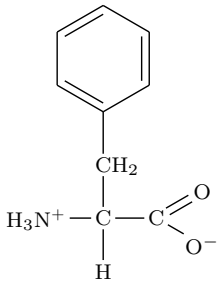
3.33 Write down the zwitterion form of the following AA:



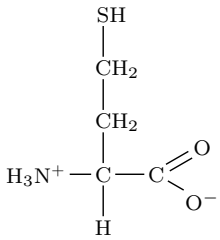
3.34 Write down the zwitterion form of the following AA:



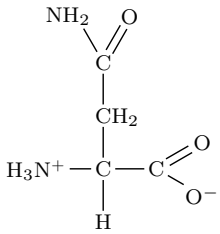
3.35 Classify the following AA as hydrophilic or hydrophobic. If hydrophilic, classify as acidic, basic or neutral:



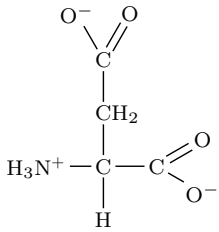
3.36 Classify the following AA as hydrophilic or hydrophobic. If hydrophilic, classify as acidic, basic or neutral:



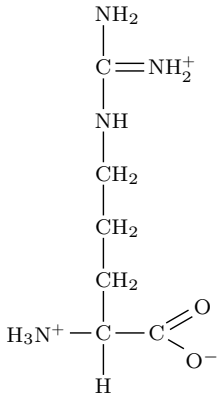
3.37 Classify the following AA as hydrophilic or hydrophobic. If hydrophilic, classify as acidic, basic or neutral:



3.38 Classify the following AA as hydrophilic or hydrophobic. If hydrophilic, classify as acidic, basic or neutral:

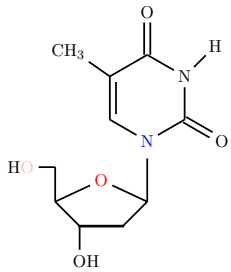


3.39 Classify the following AA as hydrophilic or hydrophobic. If hydrophilic, classify as acidic, basic or neutral:

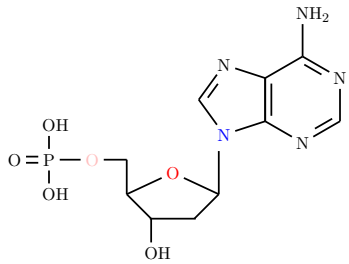


Nucleic Acids

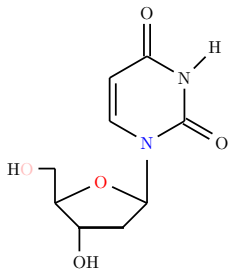
3.40 Classify the following nucleic acid as nucleotide or nucleoside.



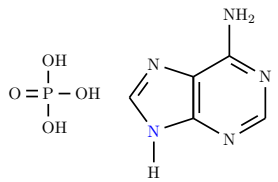
3.41 Classify the following nucleic acid as nucleotide or nucleoside.

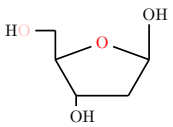


3.42 Classify the following nucleic acid as nucleotide or nucleoside.

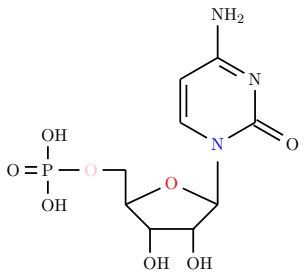


3.43 Combine the following molecules to produce a nucleoside:

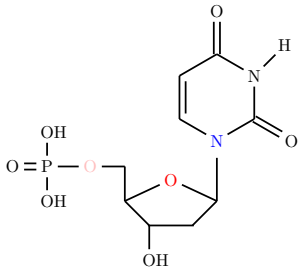




3.44 Indicate whether the following nucleic acid is part of DNA or RNA.



3.45 Indicate whether the following nucleic acid is part of DNA or RNA.



Ions & Ionic charges

3.46 Indicate if the following chemical species represent an atom, and anion or a cation: (a) Fe^{2+} (b) Cl^{-} (c) Ag

3.47 Identify the ionic state of the following elements. If needed, indicate the existence of multiple ionic states: (a) H (b) O (c) N (d) F (e) Mn

3.48 Identify the ionic state of the following elements. If needed, indicate the existence of multiple ionic states: (a) Li (b) V (c) Cl (d) S (e) Cr (f) Sr (g) Ni

Ionic compounds

3.49 Classify the following chemicals in two groups, justifying your classification: (a) NaCl (b) CO_2 (c) $FeCl_3$ (d) N_2O_3(e) SO_3(f) Ca_3N_2

3.50 Combine the following ions:

(a) $Na^{+} + Cl^{-}$
(b) $Na^{+} + Se^{2-}$
(c) $Na^{+} + P^{3-}$
(d) $Mg^{2+} + Cl^{-}$
(e) $Mg^{2+} + O^{2-}$
(f) $Mg^{2+} + N^{3-}$

3.51 Name or formulate the following ionic (Type I) compounds: (a) Magnesium iodide(b) Ca_3P_2 (c) Lithium nitride(d) MgF

3.52 Name or formulate the following ionic (Type I) compounds: (a) Magnesium fluoride(b) CaS (c) Barium phosphide(d) Mg_3N_2

3.53 Name the following compounds:

(a) NaCl
(b) Ca_3N_2
(c) MgI_2
(d) SrS
(e) RbCl
(f) KF

3.54 Combine the following ions:

(a) $Cs^{+} + Ni^{-}$
(b) $Sr^{2+} + Mn^{2-}$
(c) $Be^{2+} + Co^{4-}$
(d) $Li^{+} + Cu^{-}$
(e) $Mg^{2+} + Cr^{-6}$

ACIDS AND HYDROXIDES

3.55 Name or formulate the following acids or bases: (a) HCl (b) Hydrofluoric Acid (c) $Mg(OH)_2$

3.56 Name or formulate the following acids or bases: (a) Sulfuric Acid (b) H_2CO_3 (c) Lithium hydroxide

3.57 From the following chemicals identify acids and bases: (a) KOH (b) LiOH (c) CH_3OH

3.58 From the following chemicals identify acids and bases: (a) H_2SO_3 (b) NH_3 (c) $Ca(OH)_2$

3.59 From the following chemicals identify hydracids and oxoacids: (a) HF (b) H_2SO_3 (c) H_2S

3.60 From the following chemicals identify hydracids and oxoacids: (a) H_3BO_3 (b) HCl (c) HI

3.61 Memorize the following oxoacids:

(a) H_2SO_4 Sulfuric acid
(b) H_2CO_3 Carbonic acid
(c) $HMnO_4$ Permanganic acid
(d) HNO_3 Nitric acid
(e) H_3PO_4 Carbonic acid
(f) $H_2Cr_2O_7$ Dicromic acid

NAMING OF COMPLEX SALTS

3.62 Name or formulate the following oxoanions:

(a) ClO_4^{-}
(b) PO_4^{3-}
(c) SO_4^{2-}
(d) CO_3^{2-}
(e) NO_3^{-}
(f) CrO_4^{2-}
(g) BO_3^{3-}

3.63 Name or formulate the following (Type I) oxosalts: (a) $Mg(NO_3)_2$ (b) Sodium permanganate (c) $KMnO_4$ (d) Calcium carbonate (e) Li_3PO_4

3.64 Name or formulate the following (Type I) oxosalts: (a) Lithium sulfate (b) Na_2CrO_4 (c) Lithium sulfite (d) $Cs_2Cr_2O_7$ (e) Calcium sulfate

3.65 Name or formulate the following compounds:

(a) Na_2SO_4
(b) KNO_3
(c) $CaCO_3$
(d) $Ca(NO_2)_2$
(e) $SrSO_3$

3.66 Combine the following ions:

(a) $Na^{+} + PO_4^{3-}$
(b) $Li^{+} + MnO_4^{-}$
(c) $Mg^{2+} + NO_3^{-}$
(d) $Ca^{2+} + CO_3^{2-}$
(e) $Cs^{+} + Cr_2O_7^{2-}$
(f) $K^{+} + BO_3^{3-}$

3.67 Combine the following polyatomic ions:

(a) $Na^{+} + NO_3^{-}$
(b) $Na^{+} + CO_3^{2-}$
(c) $Na^{+} + PO_4^{3-}$
(d) $Ca^{2+} + CO_3^{2-}$
(e) $Ca^{2+} + PO_4^{3-}$

ELECTROLYTES

3.68 Indicate whether solutions of the following chemicals will have ions (I), ions and molecules (I+M), or just molecules (M):

Chemical	I	I+M	M
NaCl			
HCl			
$CaCl_2$			
H_2O			
NO_2			

USING CONVERSION FACTORS

3.69 The following conversion factor is used to convert $30cm$ into km. Fill in the gaps:

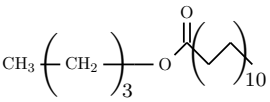
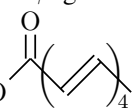
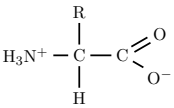
$$30\cancel{cm} \times \frac{\text{_____}\cancel{m}}{1\cancel{cm}} \times \frac{1km}{\text{_____}\cancel{m}} = 3 \times 10^{-4}km$$

3.70 The following conversion factor is used to convert $50dm$ into cm. Fill in the gaps:

$$50\cancel{dm} \times \frac{\text{_____}\cancel{m}}{1\cancel{dm}} \times \frac{1cm}{\text{_____}\cancel{m}} = 500cm$$

3.71 Fill the gap in the following conversion factors:

$$20\cancel{cm} \times \frac{1 \times 10^{-2}m}{1\cancel{cm}} \times \frac{1mm}{1 \times 10^{-3}\cancel{m}} = \text{_____}mm$$

Answers **3.1** Aldo **3.3** keto **3.5** α **3.7** β **3.9** d **3.11** d **3.13** α-1-4-glycosidic bond **3.15** β-1-4-glycosidic bond **3.17** β-galactose and α-glucose **3.19** β **3.23** monounsaturated **3.25** $C_8H_{12}O_2$ **3.27**  **3.29**  **3.31** left **3.33**  **3.35** hydrophobic **3.37** hydrophilic neutral **3.39** hydrophilic basic **3.41** Nucleotide **3.43** Not given **3.45** DNA **3.47** (a) H^+ (b) O^{2-} (c) N^{3-} (d) F^- (e) Mn (multiple) **3.49** (a) NaCl (group ionic) (b) CO_2 (group covalent) (c) $FeCl_3$ (group ionic) (d) N_2O_3 (group covalent) (e) SO_3 (group covalent) (f) Ca_3N_2 (group ionic) **3.51** (a) Magnesium iodide (MgI_2) (b) Ca_3P_2 (Calcium phosphide) (c) Lithium nitride (Li_3N) (d) MgF (Magnesium fluoride) **3.53** (a) NaCl (sodium chloride) (b) Ca_3N_2 (calcium nitride) (c) MgI_2 (magnesium iodide) (d) SrS (strontium sulfide) (e) RbCl (rubidium chloride) (f) KF (potassium fluoride) **3.55** (a) HCl (hydrochloric acid) (b) Hydrofluoric Acid (HF) (c) $Mg(OH)_2$ (magnesium hydroxide) **3.57** (a) KOH (base) (b) LiOH (base) (c) CH_3OH (acid, organic) **3.59** (a) HF (hydracid) (b) H_2SO_3 (oxacid) (c) H_2S (hydracid) **3.61** No answer. **3.63** (a) $Mg(NO_3)_2$ (magnesium nitrate) (b) Sodium permanganate ($NaMnO_4$) (c) $KMnO_4$ (potassium permanganate) (d) Calcium carbonate ($CaCO_3$) (e) Li_3PO_4 (lithium phosphate) **3.64** (a) Lithium sulfate (Li_2SO_4) (b) Na_2CrO_4 (sodium dichromate) (c) Lithium sulfite (Li2SO3) (d) $Cs_2Cr_2O_7$ (caesium dichromate) (e) Calcium sulfate (CaSO4) **3.69** $1 \cdot 10^{-2}; 10 \cdot 10^3$